

Foundations of Probability, Random Walks and Diffusion

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Mathematical Methods for Physicists II

Coursework Notes and Computational Guide

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This is a collection of notes compiled from my Mathematical Methods for Physicists II coursework on probability, random walks and diffusion, with explanatory material supporting the accompanying computational exercises. It introduces the concepts and mathematical reasoning needed to develop the logic of a simulation or numerical solution. These are learning notes rather than a comprehensive textbook treatment.

1 Experiments, outcomes and probability

A random experiment has several possible outcomes, but its next outcome is not known in advance. The **sample space** is the set of possible outcomes. For one coin toss,

$$\Omega = \{\text{heads, tails}\}.$$

An **event** is a collection of outcomes. For one toss, “heads” is an event; for several tosses, “exactly two heads” is another event. Probability assigns a number to an event:

$$0 \leq P(A) \leq 1.$$

Probabilities of mutually exclusive outcomes covering all possibilities sum to one. For a fair coin, $P(H) = P(T) = 1/2$.

1.1 Relative frequency

If an event occurs C times in N trials, its observed relative frequency is

$$\hat{P} = \frac{C}{N}.$$

A hat denotes an estimate rather than an exact theoretical value. Observing 53 heads in 100 tosses gives $\hat{P}(H) = 0.53$; a finite experiment need not reproduce the model probability exactly. To estimate probability computationally, repeat the experiment, count outcomes satisfying the event, and divide by the trial number. Store a history if the evolution of the estimate is needed.

2 Random variables

A random variable assigns a numerical value to each outcome. A Bernoulli variable is

$$B = \begin{cases} 1, & \text{with probability } q, \\ 0, & \text{with probability } 1 - q. \end{cases}$$

For a walk, use the displacement variable

$$\xi = \begin{cases} +a, & \text{rightward step,} \\ -a, & \text{leftward step,} \end{cases}$$

where $a > 0$ is the step length. A 0/1 representation makes counting convenient; a $-a/+a$ representation makes position updates convenient.

3 Expectation, variance and standard deviation

3.1 Expectation

For a discrete variable,

$$E[X] = \sum_x xP(X = x).$$

Expectation is a probability-weighted average, not necessarily a possible individual outcome. For a fair step, $E[\xi] = a/2 - a/2 = 0$, although no individual step is zero. Expectation is linear:

$$E[X + Y] = E[X] + E[Y],$$

without requiring independence.

3.2 Variance

Variance measures spread about the mean:

$$\text{Var}(X) = E[(X - E[X])^2] = E[X^2] - E[X]^2.$$

Standard deviation is $\sigma_X = \sqrt{\text{Var}(X)}$. If position has length units, variance has length-squared units and standard deviation has length units.

3.3 Estimation from samples

For M sampled values,

$$\bar{x} = \frac{1}{M} \sum_{m=1}^M x_m, \quad s_M^2 = \frac{1}{M} \sum_{m=1}^M (x_m - \bar{x})^2.$$

The exact sample identity

$$\frac{1}{M} \sum_m x_m^2 = s_M^2 + \bar{x}^2$$

distinguishes the mean of squares from the variance about the sample mean.

4 Independence and correlation

Independent variables satisfy $E[XY] = E[X]E[Y]$. Their variances add. More generally,

$$\text{Var}(X + Y) = \text{Var}(X) + \text{Var}(Y) + 2 \text{Cov}(X, Y).$$

Covariance measures how variables fluctuate together. In an unrestricted walk, successive *increments* are independent, but successive *positions* are correlated because each contains earlier increments. This determines what counts as an independent sample.

5 Coin counts and the binomial distribution

Let K count successes in N independent trials with success probability q . A particular sequence containing k successes has probability $q^k(1-q)^{N-k}$. There are $\binom{N}{k} = N!/[k!(N-k)!]$ such sequences, so

$$P(K = k) = \binom{N}{k} q^k (1-q)^{N-k}.$$

Its moments are

$$E[K] = Nq, \quad \text{Var}(K) = Nq(1-q).$$

For the observed success fraction,

$$E[K/N] = q, \quad \text{Std}(K/N) = \sqrt{\frac{q(1-q)}{N}}.$$

Typical fractional fluctuations decrease as $N^{-1/2}$. The law of large numbers describes convergence of this fraction to q , without requiring monotonic improvement along a sequence.

6 Random Walk

A random walk updates a state with a random increment:

$$X_{n+1} = X_n + \xi_{n+1}.$$

Starting at $X_0 = 0$,

$$X_N = \sum_{j=1}^N \xi_j.$$

For the fair walk, $P(\xi_j = +a) = P(\xi_j = -a) = 1/2$. A stochastic process is a family of random variables indexed by time or step count. This walk has the Markov property: given the current position, the next-position probabilities do not require the earlier history.

6.1 Position, displacement and distance

Position is X_N , displacement is $X_N - X_0$, and total distance travelled is $\sum_j |\xi_j|$. An unrestricted fixed-length walk travels distance Na even when opposite steps leave a small final displacement.

7 Spreading of a fair walk

For independent fair increments, $E[\xi_j] = 0$ and $E[\xi_j^2] = a^2$, so $E[X_N] = 0$. Expand the square:

$$E[X_N^2] = \sum_j E[\xi_j^2] + 2 \sum_{i < j} E[\xi_i \xi_j].$$

Independence makes each cross term $E[\xi_i]E[\xi_j] = 0$. Therefore,

$$E[X_N^2] = Na^2, \quad \text{Var}(X_N) = Na^2, \quad \sigma_N = a\sqrt{N}.$$

A cloud can have zero mean while its spread increases. These are ensemble predictions, not a claim that every walker stays at the origin.

8 Exact endpoints, parity and Gaussian approximation

If K steps go right and $N - K$ go left,

$$X_N = a(2K - N).$$

The binomial distribution of K gives the exact endpoint probabilities. For an integer m ,

$$P(X_N = ma) = \binom{N}{(N+m)/2} q^{(N+m)/2} (1-q)^{(N-m)/2},$$

provided $|m| \leq N$ and $N + m$ is even; otherwise the probability is zero. Even step counts reach even multiples of a ; odd counts reach odd multiples. Accessible sites at fixed age are separated by $2a$. The central limit theorem states that a properly centred and scaled sum of many independent, identically distributed increments with finite nonzero variance approaches a Gaussian. For fair steps, the large- N envelope has mean zero and variance Na^2 . Its smooth shape does not remove discrete parity.

9 Trajectories, ensembles and time occupancy

A trajectory is one history $X_0, \dots, X_N$. An ensemble comprises independent histories under the same rule. An ensemble average at fixed step n is

$$\widehat{E[X_n]} = \frac{1}{M} \sum_{m=1}^M X_n^{(m)}.$$

Time occupancy instead counts positions along one history:

$$\hat{\mu}_N(x) = \frac{1}{N} \sum_{n=1}^N \mathbf{1}_{\{X_n=x\}}.$$

The indicator equals one when the walker occupies x . Occupancy answers where one walker spent its observed time; endpoint probabilities answer where independent walkers finish at a fixed age. They cannot be interchanged. The unrestricted fair walk on an infinite line has no normalizable stationary position distribution.

10 Probability mass, density and cumulative distributions

A discrete mass function satisfies $\sum_j P(X = x_j) = 1$. A continuous density gives probabilities through integrals:

$$P(x_1 \leq X \leq x_2) = \int_{x_1}^{x_2} p(x) dx, \quad \int_{-\infty}^{\infty} p(x) dx = 1.$$

Density has units of inverse length, is not a probability at a point, and can exceed one. The CDF is $F(x) = P(X \leq x)$: a running sum for discrete masses or an integral for a continuous density.

10.1 Histogram normalization

With M observations and bin width w , density height is count/ (Mw) . Total bar area is one when all data are included. To compare a fixed-age lattice walk with a smooth density, account for accessible-site spacing $2a$.

11 Two-dimensional walks and boundaries

A cardinal walk chooses $(+a, 0)$, $(-a, 0)$, $(0, +a)$ or $(0, -a)$ equally. Only one coordinate moves per tick:

$$E[X_N^2] = E[Y_N^2] = Na^2/2, \quad E[R_N^2] = Na^2.$$

Boundary rules change the process. Rejection holds the walker at its current site; absorption permits probability to leave; periodic boundaries connect opposite edges. For the finite square's rejection rule, neighboring transitions have equal forward and reverse probabilities. Uniform site probabilities satisfy detailed balance and form the stationary distribution. Confinement prevents indefinite spreading; a finite trajectory may not yet sample equilibrium evenly.

12 Diffusion and physical time

Let τ be time per particle step and $t = N\tau$. The fair one-dimensional variance is $Na^2 = (a^2/\tau)t$. Matching $\text{Var}(X) = 2Dt$ identifies

$$D = \frac{a^2}{2\tau}.$$

D has units of length squared/time. The two-dimensional cardinal walk instead has $E[R^2] = 4Dt$, giving $D = a^2/(4\tau)$ for that microscopic rule.

13 From lattice motion to the diffusion equation

Let P_j^n be probability at site j after n steps. Fair arrival from either neighbor gives

$$P_j^{n+1} = \frac{1}{2}P_{j-1}^n + \frac{1}{2}P_{j+1}^n.$$

A smooth coarse-grained density obeys approximately

$$p(x, t + \tau) \approx \frac{1}{2}p(x - a, t) + \frac{1}{2}p(x + a, t).$$

Taylor-expand:

$$p(x, t + \tau) = p + \tau p_t + \dots, \quad p(x \pm a, t) = p \pm ap_x + \frac{1}{2}a^2 p_{xx} + \dots.$$

Odd spatial terms cancel. Retaining leading terms yields $\tau p_t = (a^2/2)p_{xx}$ and

$$p_t = Dp_{xx}.$$

This describes scales large compared with microscopic steps. A controlled diffusion limit sends a, τ to zero with a^2/τ fixed.

13.1 Flux interpretation

Probability conservation is $p_t + \partial_x J = 0$. Fick's law $J = -D\partial_x p$ states that probability flows down its density gradient. Together they give diffusion.

14 The point-release solution

A unit point release is $p(x, 0) = \delta(x)$. On an infinite line with constant D ,

$$p(x, t) = \frac{e^{-x^2/(4Dt)}}{\sqrt{4\pi Dt}}, \quad t > 0.$$

Its integral is one, mean zero, and variance $2Dt$. Fourier transformation gives $d\hat{p}/dt = -Dk^2\hat{p}$. The delta initial condition has transform one, so $\hat{p} = e^{-Dk^2t}$; inversion yields the Gaussian. At $t = 0$, use the delta initial condition rather than the singular formula.

15 Numerical derivatives and the grid solution

Grid sites have spacing Δx and numerical times spacing Δt . Approximate

$$p_t \approx \frac{p_i^{n+1} - p_i^n}{\Delta t}, \quad p_{xx} \approx \frac{p_{i+1}^n - 2p_i^n + p_{i-1}^n}{(\Delta x)^2}.$$

The spatial formula follows by adding Taylor expansions at the two neighbors. Substitution gives

$$p_i^{n+1} = p_i^n + r(p_{i+1}^n - 2p_i^n + p_{i-1}^n), \quad r = \frac{D\Delta t}{(\Delta x)^2}.$$

All right-hand values are from the old time level. A separate new array prevents mixing time levels.

15.1 Stability and accuracy

A Fourier mode has amplification $G = 1 - 4r \sin^2(k\Delta x/2)$. Requiring $|G| \leq 1$ for all modes gives $0 \leq r \leq 1/2$. In that range,

$$p_i^{n+1} = (1 - 2r)p_i^n + rp_{i-1}^n + rp_{i+1}^n$$

is also a nonnegative weighted average. Stability controls error growth; accuracy additionally requires suitable grid resolution and refinement checks.

16 Bias and drift

For right-step probability q ,

$$E[\xi] = a(2q - 1), \quad \text{Var}(\xi) = 4a^2q(1 - q).$$

After N independent steps,

$$E[X_N] = Na(2q - 1), \quad \text{Var}(X_N) = 4Na^2q(1 - q).$$

Define $v = a(2q - 1)/\tau$ and $D_{\text{spread}} = 2a^2q(1 - q)/\tau$. The mean is vt and variance $2D_{\text{spread}}t$. A large-step-count Gaussian approximation is described by

$$p_t = -vp_x + D_{\text{spread}}p_{xx}.$$

Drift moves the centre; diffusion broadens the cloud around it. This approximation matches lattice moments, not its exact discrete evolution. A finite- v , finite- D continuum limit also requires appropriate scaling of the bias.

17 Turning a model into a program

Before coding, identify the state, initial condition, update rule, meaning of one iteration, stored quantities, boundaries and theoretical checks.

Question	Example
State	One position, a particle array, or a density array
Initial condition	All particles at zero
Update	A random increment or a finite-difference step
Iteration	A particle time step or a numerical time step
Storage	Full paths, snapshots, or summary statistics
Boundaries	Unrestricted, rejecting, or absorbing
Checks	Normalization, predicted moments, nonnegative density

A useful sequence is: initialize, update, record, repeat, analyse, compare with theory. A fixed seed supports reproducibility but does not remove sampling uncertainty. Draw order can affect later results.